Министерство образования Республики Беларусь Учреждение образования
«Брестский государственный технический университет»
Кафедра ИИТ

Лабораторная работа №3
По дисциплине: «интеллектуальный анализ данных»
По теме: «**Предобучение нейронных сетей с использованием автоэнкодерного подхода**»

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Брест 2026

Цель: научиться осуществлять предобучение нейронных сетей с помощью автоэнкодерного подхода.

| № | Выборка | Тип задачи | Целевая переменная |
|---|---|---|---|
| 5 | https://archive.ics.uci.edu/dataset/193/cardiotocography | классификация | CLASS/NSP |

**Результаты работы сети:**

```
Датасет: cardiotocography
Размер: (2126, 21)
Количество классов: 3

Обучение без предобучения...
without_pretraining: эпоха 10, train loss=0.2266, valid loss=0.2465, valid accuracy=0.9028
without_pretraining: эпоха 20, train loss=0.1631, valid loss=0.2189, valid accuracy=0.9216
without_pretraining: эпоха 30, train loss=0.1268, valid loss=0.2161, valid accuracy=0.9216
without_pretraining: эпоха 40, train loss=0.1009, valid loss=0.2131, valid accuracy=0.9279
without_pretraining: эпоха 50, train loss=0.0861, valid loss=0.2332, valid accuracy=0.9248
without_pretraining: эпоха 60, train loss=0.0701, valid loss=0.2419, valid accuracy=0.9248
without_pretraining: эпоха 70, train loss=0.0628, valid loss=0.2650, valid accuracy=0.9248
without_pretraining: эпоха 80, train loss=0.0523, valid loss=0.2711, valid accuracy=0.9310
without_pretraining: эпоха 90, train loss=0.0439, valid loss=0.2998, valid accuracy=0.9185
without_pretraining: эпоха 100, train loss=0.0368, valid loss=0.3109, valid accuracy=0.9154

without_pretraining
Accuracy: 0.8966
Macro F1: 0.7845
              precision    recall  f1-score   support

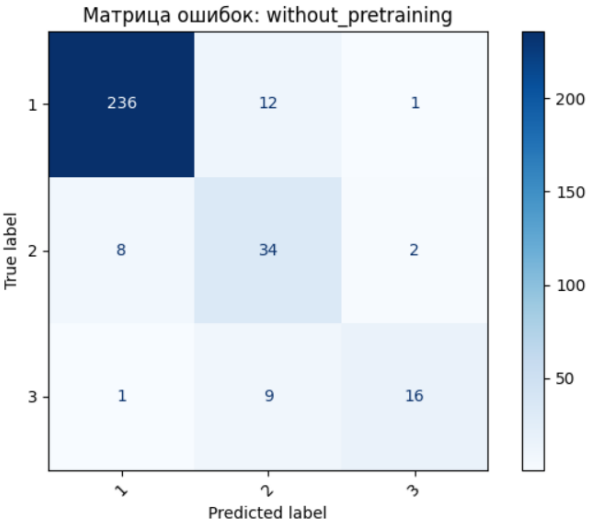
           1       0.96      0.95      0.96       249
           2       0.62      0.77      0.69        44
           3       0.84      0.62      0.71        26

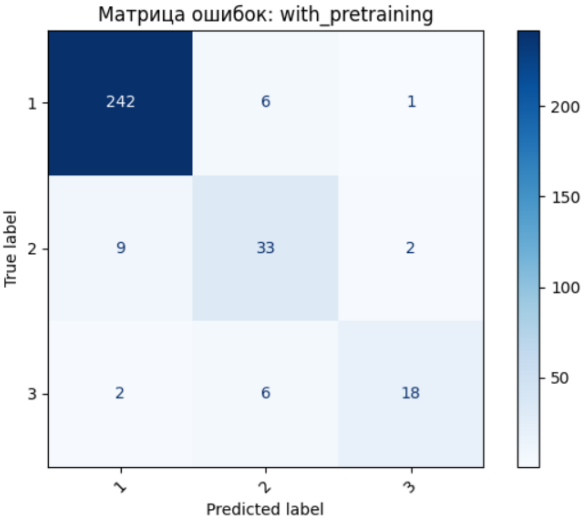
    accuracy                           0.90       319
   macro avg       0.81      0.78      0.78       319
weighted avg       0.91      0.90      0.90       319
Обучение с предобучением...
Предобучение первого скрытого слоя...
  Предобучение: эпоха 10, MSE = 0.07286
  Предобучение: эпоха 20, MSE = 0.01854
  Предобучение: эпоха 30, MSE = 0.01056
Предобучение второго скрытого слоя...
  Предобучение: эпоха 10, MSE = 0.05817
  Предобучение: эпоха 20, MSE = 0.02541
  Предобучение: эпоха 30, MSE = 0.01734
with_pretraining: эпоха 10, train loss=0.2318, valid loss=0.2517, valid accuracy=0.9060
with_pretraining: эпоха 20, train loss=0.1782, valid loss=0.2269, valid accuracy=0.9216
with_pretraining: эпоха 30, train loss=0.1497, valid loss=0.2199, valid accuracy=0.9185
with_pretraining: эпоха 40, train loss=0.1234, valid loss=0.2231, valid accuracy=0.9060
with_pretraining: эпоха 50, train loss=0.1073, valid loss=0.2111, valid accuracy=0.9310
with_pretraining: эпоха 60, train loss=0.0908, valid loss=0.2195, valid accuracy=0.9154
with_pretraining: эпоха 70, train loss=0.0798, valid loss=0.2297, valid accuracy=0.9060
with_pretraining: эпоха 80, train loss=0.0741, valid loss=0.2463, valid accuracy=0.9028
with_pretraining: эпоха 90, train loss=0.0621, valid loss=0.2514, valid accuracy=0.9185
with_pretraining: эпоха 100, train loss=0.0591, valid loss=0.2693, valid accuracy=0.8997

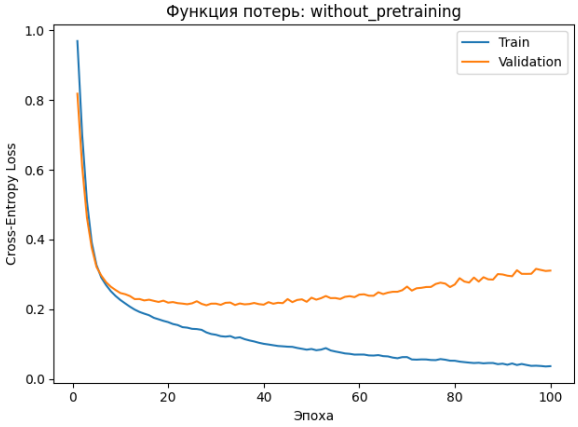
with_pretraining
Accuracy: 0.9185
Macro F1: 0.8239
              precision    recall  f1-score   support

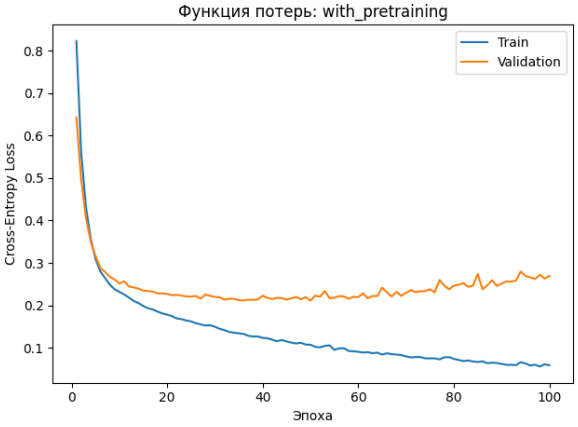
           1       0.96      0.97      0.96       249
           2       0.73      0.75      0.74        44
           3       0.86      0.69      0.77        26

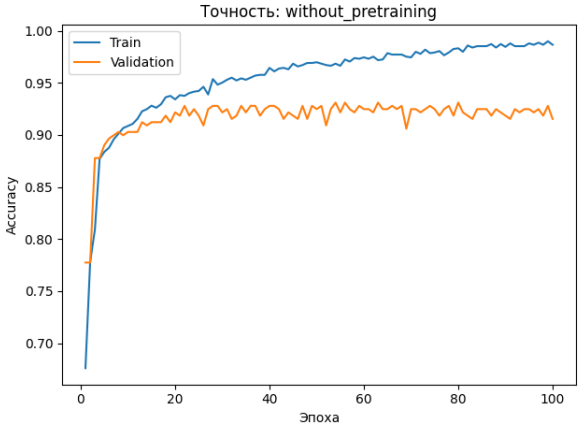
    accuracy                           0.92       319
   macro avg       0.85      0.80      0.82       319
weighted avg       0.92      0.92      0.92       319
```

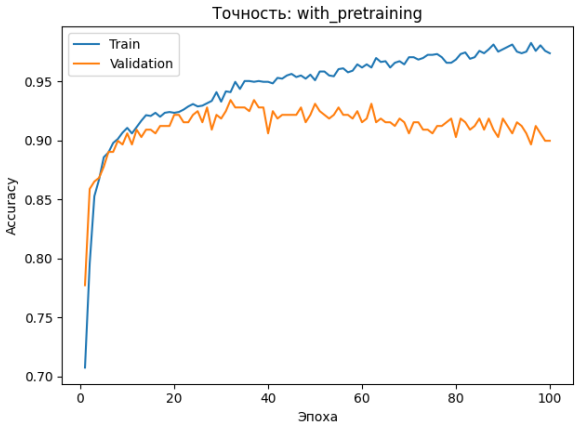

Матрица ошибок: without_pretraining
236
12
1
8
34
2
1
9
16
True label
Predicted label
1
2
3
200
150
100
50


Матрица ошибок: with_pretraining
242
6
1
9
33
2
2
6
18
True label
Predicted label
1
2
3
200
150
100
50


Функция потерь: without_pretraining
Train
Validation
Cross-Entropy Loss
Эпоха


Функция потерь: with_pretraining
Train
Validation
Cross-Entropy Loss
Эпоха


Точность: without_pretraining
Train
Validation
Accuracy
Эпоха


Точность: with_pretraining
Train
Validation
Accuracy
Эпоха

```
Датасет: digits
Размер: (1797, 64)
Количество классов: 10

Обучение без предобучения...
without_pretraining: эпоха 10, train loss=0.1486, valid loss=0.2083, valid accuracy=0.9333
without_pretraining: эпоха 20, train loss=0.0319, valid loss=0.1456, valid accuracy=0.9593
without_pretraining: эпоха 30, train loss=0.0114, valid loss=0.1356, valid accuracy=0.9593
without_pretraining: эпоха 40, train loss=0.0057, valid loss=0.1352, valid accuracy=0.9593
without_pretraining: эпоха 50, train loss=0.0034, valid loss=0.1378, valid accuracy=0.9630
without_pretraining: эпоха 60, train loss=0.0022, valid loss=0.1391, valid accuracy=0.9630
without_pretraining: эпоха 70, train loss=0.0016, valid loss=0.1405, valid accuracy=0.9630
without_pretraining: эпоха 80, train loss=0.0012, valid loss=0.1426, valid accuracy=0.9630
without_pretraining: эпоха 90, train loss=0.0009, valid loss=0.1439, valid accuracy=0.9630
without_pretraining: эпоха 100, train loss=0.0007, valid loss=0.1461, valid accuracy=0.9630

without_pretraining
Accuracy: 0.9852
Macro F1: 0.9848
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        27
           1       0.93      1.00      0.97        28
           2       0.96      1.00      0.98        26
           3       1.00      1.00      1.00        28
           4       1.00      1.00      1.00        27
           5       1.00      1.00      1.00        27
           6       1.00      0.96      0.98        27
           7       1.00      1.00      1.00        27
           8       0.96      0.88      0.92        26
           9       1.00      1.00      1.00        27

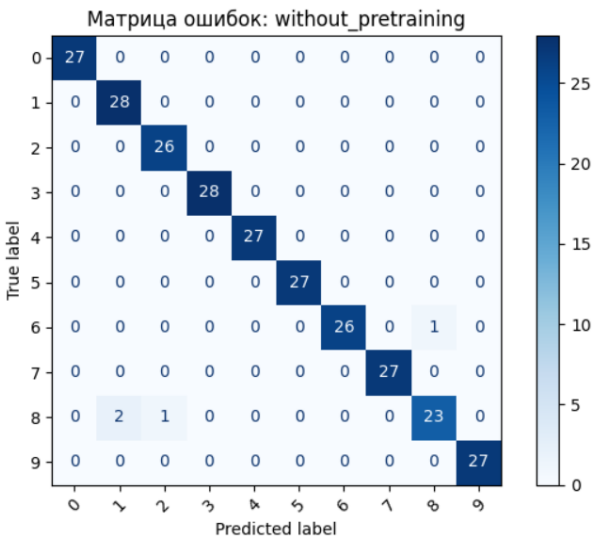
    accuracy                           0.99       270
   macro avg       0.99      0.98      0.98       270
weighted avg       0.99      0.99      0.99       270

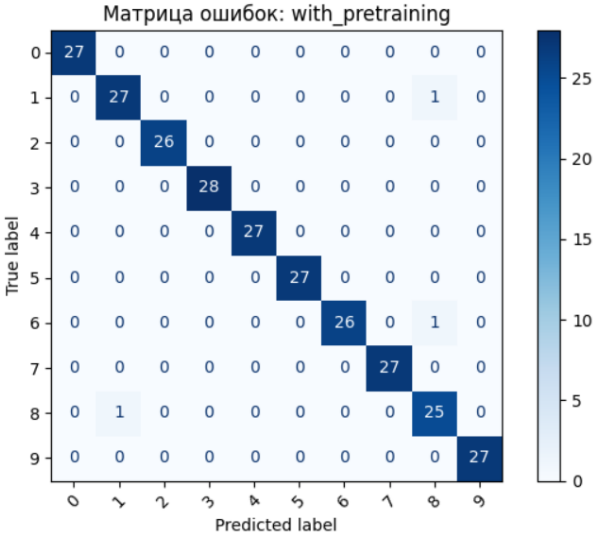
Обучение с предобучением...
Предобучение первого скрытого слоя...
  Предобучение: эпоха 10, MSE = 0.21787
  Предобучение: эпоха 20, MSE = 0.09039
  Предобучение: эпоха 30, MSE = 0.05241
Предобучение второго скрытого слоя...
  Предобучение: эпоха 10, MSE = 0.17494
  Предобучение: эпоха 20, MSE = 0.09018
  Предобучение: эпоха 30, MSE = 0.06168
with_pretraining: эпоха 10, train loss=0.1665, valid loss=0.2228, valid accuracy=0.9481
with_pretraining: эпоха 20, train loss=0.0493, valid loss=0.1556, valid accuracy=0.9630
with_pretraining: эпоха 30, train loss=0.0207, valid loss=0.1443, valid accuracy=0.9630
with_pretraining: эпоха 40, train loss=0.0105, valid loss=0.1420, valid accuracy=0.9667
with_pretraining: эпоха 50, train loss=0.0062, valid loss=0.1447, valid accuracy=0.9667
with_pretraining: эпоха 60, train loss=0.0041, valid loss=0.1456, valid accuracy=0.9667
with_pretraining: эпоха 70, train loss=0.0028, valid loss=0.1498, valid accuracy=0.9704
with_pretraining: эпоха 80, train loss=0.0021, valid loss=0.1505, valid accuracy=0.9704
with_pretraining: эпоха 90, train loss=0.0016, valid loss=0.1532, valid accuracy=0.9704
with_pretraining: эпоха 100, train loss=0.0013, valid loss=0.1540, valid accuracy=0.9704

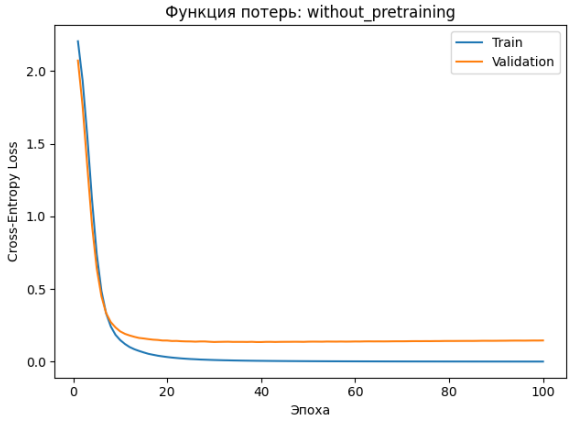
with_pretraining
Accuracy: 0.9889
Macro F1: 0.9889
              precision    recall  f1-score   support

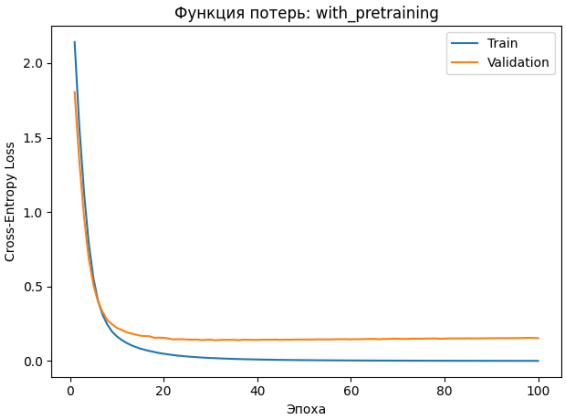
           0       1.00      1.00      1.00        27
           1       0.96      0.96      0.96        28
           2       1.00      1.00      1.00        26
           3       1.00      1.00      1.00        28
           4       1.00      1.00      1.00        27
           5       1.00      1.00      1.00        27
           6       1.00      0.96      0.98        27
           7       1.00      1.00      1.00        27
           8       0.93      0.96      0.94        26
           9       1.00      1.00      1.00        27

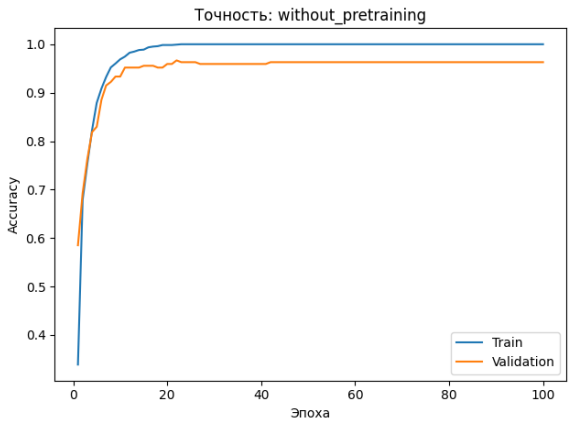
    accuracy                           0.99       270
   macro avg       0.99      0.99      0.99       270
weighted avg       0.99      0.99      0.99       270
```

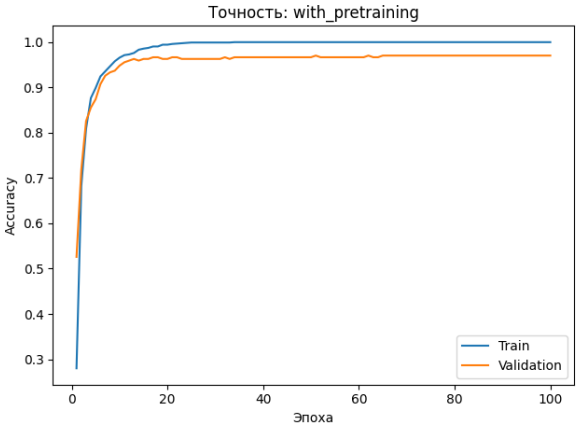



```
Итоговое сравнение:
                model  accuracy  macro_f1
without_pretraining  0.985185  0.984778
   with_pretraining  0.988889  0.988881
without_pretraining  0.896552  0.784482
   with_pretraining  0.918495  0.823891
```

**Вывод:** научился осуществлять предобучение нейронных сетей с помощью автоэнкодерного подхода.